



These vacuum tubes will carry specially designed pods at uber-fast speeds



An artist's imagination of Hyperloop stations

Mumbai and Pune, incidentally, have also been identified as potential candidates for smart cities. It is important for smart cities to have their own identity when it comes to transportation - which helps the city build upon its mobility options. Alongside, Ford enumerated the need for efficient, non-polluting and fast modes of shared transportation that solves congestion. Taking these into account, Hyperloop is the perfect candidate.

At present, the average travel time between the two cities, taking into account the average congestion, can go up to almost four hours. With companies pushing for smart last mile connectivity, Hyperloop can serve as the ideal mass transit system that can get people from Mumbai to Pune (and vice versa) in less than 30 minutes. On average, the overall point-to-point travel time can go down to as little as one hour. This in turn would ease congestion on roadways, leave a wider choice of fast mobility options. Hyperloop can be the ideal push towards smarter livelihood.

The stats: Possible cost and viability

When Musk first floated the Hyperloop concept, he quoted an estimated

figure Rs. 38,500 crore. However, in late-2016, it was revealed that the cost of building and running Hyperloop could go up to as much as Rs. 84,260 crore.

In terms of cost, the Mumbai-Pune corridor in India would be advantageous over other developed nations. The cost overall would be significantly lower between Mumbai and Pune, than in Los Angeles or Dubai.

VHO (Virgin Hyperloop One) has estimated a total cost of about Rs. 20,000 crore to host the feasibility tests, establish the Hyperloop network between Mumbai and Pune, which is a meagre fraction of its possible counterpart in LA and Dubai. VHO chief Branson has further stated that the proposed Hyperloop route will have the capacity to ferry 150 million passengers every year. At approximately Rs. 400 per passenger at its proposed maximum capacity, the Hyperloop project may even incur profits within five operational years.

In comparison, the proposed bullet train route between Mumbai and Ahmedabad has been quoted at an estimated expenditure of a staggering 1,10,160 crore. This route would also require additional supply of electricity to run and specialised platform

hubs, and would not be a fast mode of transportation in comparison to what Hyperloop can potentially achieve. This route would be more expensive to maintain, be slower in commute duration, and be a sizeable expenditure to the government's costs.

The future

While the feasibility tests are yet to be undertaken, Hyperloop makes an immense amount of sense in India. It takes lesser ground to be established, generates its own power, drastically reduces carbon and greenhouse gas emissions, run at significantly lesser cost per customer, establish a ground for Mumbai and Pune's quest to evolve as smart cities, and above all, bring the travelling time down by almost 600 percent.

Many may argue that air travel is already in place for fast commuting, but once all factors are taken into account, it is barely a fast procedure, and for the roughly 250km stretch of Mumbai and Pune, the total air travel time exceeds well above four hours from point of origin to destination. It is this, and much more, that makes Hyperloop an ideal candidate for this corridor, and multiple other similar routes all across India. ■